

Required and optional software for the shortcourse



- <https://julialang.org/downloads/>
 - Allow to install and use MAGEMinApp and MAGEMin_C
- Visual Studio Code – Code editor <https://code.visualstudio.com/>
 - Allow Windows user to connect to Linux WSL and get the best performance for MAGEMin
- <https://inkscape.org/fr/>
 - Allow to open MAGEMinApp figure vector format (svg)
 - Illustrator works too...



- **WINDOWS users**
- Recommended to install and use WSL (Windows sub-system for Linux)
- Connect to WSL and Linux Via VSCode (see next slides)
- MAGEMin performs twice faster on Linux/Mac than native Windows

Julia installation

**Windows,
Mac and Linux**

Julia installation - <https://julialang.org/downloads/>

Windows

Install julia

Install the latest Julia version ([v1.11.4](#) March 10, 2025) from the [Microsoft Store](#) by running this in the command prompt:

```
> winget install --name Julia --id 9NJNWW8PVKMN -e -s msstore
```

Powershell

COPY

For Linux and MacOS instructions [click here](#)

Once installed **julia** will be available via the command line interface.

This will install the [Juliaup](#) installation manager, which will automatically install julia and help keep it up to date. The command **juliaup** is also installed. To install different julia versions see **juliaup --help**.

Please star us [on GitHub](#). If you use Julia in your research, please [cite us](#). If possible, do consider [sponsoring](#) us.

Install julia

Install the latest Julia version ([v1.11.4](#) March 10, 2025) by running this in your terminal:

```
$ curl -fsSL https://install.julialang.org | sh
```

Terminal

COPY

It looks like you're using a Unix/Linux-type system. For Windows instructions [click here](#)

Once installed **julia** will be available via the command line interface.

This will install the [Juliaup](#) installation manager, which will automatically install julia and help keep it up to date. The command **juliaup** is also installed. To install different julia versions see **juliaup --help**.

Please star us [on GitHub](#). If you use Julia in your research, please [cite us](#). If possible, do consider [sponsoring](#) us.

Linux/Mac

MAGEMinApp & MAGEMin_C installation

Windows
Mac and Linux

Install MAGEMinApp

- Open a new terminal (vscode for those who use it, or terminal/powershell for others)
- Type:
julia -t 6 # where 6 is the number of core you want to use (depends on your machine, type
 # 'versioninfo()' to get more informations)

```
○ seph42@DESKTOP-2V82075:~$  
○ seph42@DESKTOP-2V82075:~$ julia -t 6  
  
┌───┐ ┌───┐ ┌───┐ | Documentation: https://docs.julia.org  
│   │ │   │ │   │ |  
│   │ │   │ │   │ | Type "?" for help, "]?" for Pkg help.  
├───┴───┘ └───┬───┘ |  
│               │   │ | Version 1.10.0 (2023-12-25)  
│               │   │ | Official https://julia.org/ release  
└───┬───┘ └───┬───┘ |  
     │         │  
     └────────┘
```

(v1.10) pkg> add MAGERMinApp

- In the terminal type ']', this will open the package manager
- Type 'add MAGEMinApp', this will download and install MAGEMinApp
- Once installed, quite the package manager by typing 'BACKSPACE' key

Launch MAGEMinApp

- In the Julia terminal, type:
‘using MAGEMinApp’
Then
‘App()’
- The following text will be displayed in the terminal:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
seph42@DESKTOP-2V82075:~$ julia -t 6

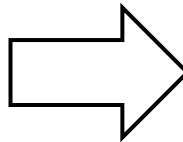
Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.

Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

(julia) pkg> add MAGEMinApp
Resolving package versions...
No Changes to `~/julia/environments/v1.10/Project.toml`
No Changes to `~/julia/environments/v1.10/Manifest.toml`

julia> using MAGEMinApp
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 1
```



- Copy and paste the address in your web-browser:

erc JGU UNIVERSITÄT GUTENBERG REMINDER: Don't forget to properly cite the references used to create the diagrams by using the 'export references' button!

MAGEMin

File

Phase diagrams PTX path Isotropic path

Setup Diagram Trace elements

Phase diagram parameters

Thermodynamic database: Ignavea (Holland)

Solution phase selection: ☐

Diagram type: P-T diagram

☐ Limit Ca-ops: 0.5

TE predictive model: false

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 800 max 1400

Initial grid subdivision: 3

Refinement type: Phases only

Refinement levels: 2

Buffer: no buffer

Solver: Hybrid

Verbose: none

Specific Cp: G0

Bulk-rock composition

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	mol_fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

ig

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

Compute phase diagram

id=b52cd012-7d16-4c5f-8d85-ce0d07c153a, MAGEMinApp GUI v=0.3.4

MAGEMin_C, the julia interface: installation



https://github.com/ComputationalThermodynamics/MAGEMin_C.jl

Install MAGEMin_C

```
julia> ] # opens the package
manager
pkg> add MAGEMin_C # MAGEMin_C
```

Load MAGEMin_C

```
julia> using MAGEMin_C # load MAGEMin_C
```

```
seph42@DESKTOP-2V82075:~$ julia

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

(@v1.10) pkg> add MAGEMin_C
Resolving package versions...
Updating `~/.julia/environments/v1.10/Project.toml`
[e5d170eb] + MAGEMin_C v1.5.5
No Changes to `~/.julia/environments/v1.10/Manifest.toml`

julia> using MAGEMin_C
Using libMAGEMin.dylib from MAGEMin_jll
```